Beyond Static Benchmarks: A Post-Leaderboard Evaluation Paradigm for Generative Music and Creative AI

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TL;DR

A *community-governed*, *multi-dimensional* evaluation ecosystem for music AI replacing static leaderboards with transparent, reproducible *benchmark profiles*.

Problem: Evaluation Crisis

Generative music evaluation is **methodologically fragmented**, relies on **ambiguous validation objectives**, and relies on **misaligned metrics** [1].

- **Validity failure:** metrics poorly capture musical qualities.
- **Leaderboard failure:** single scores encourage optimization artifacts and obscure musical multi-dimensionality.
- **Comparability failure:** different evaluation pipelines yield incomparable results.
- **Reproducibility failure:** proprietary data and tools limit verification.
- **Governance failure:** no shared standards or process for evolution.

Core Ideas

- **Community governance:** *community-maintained evolving benchmarks* and evaluation standards.
- **Multi-dimensional assessment:** capture many *relevant dimensions* instead of a single ambiguous score.
- **Interpretability:** utilize *interpretable metrics* and features.
- **Use-case adaptability:** tailor curated *reference benchmark profiles (RBPs)* to specific musical contexts.
- **Modular, extensible architecture:** new metrics, features, and data can be *integrated without redesign*.
- **Reproducibility & transparency:** versioned, documented *open source* software with *auditable logging*.

Governance

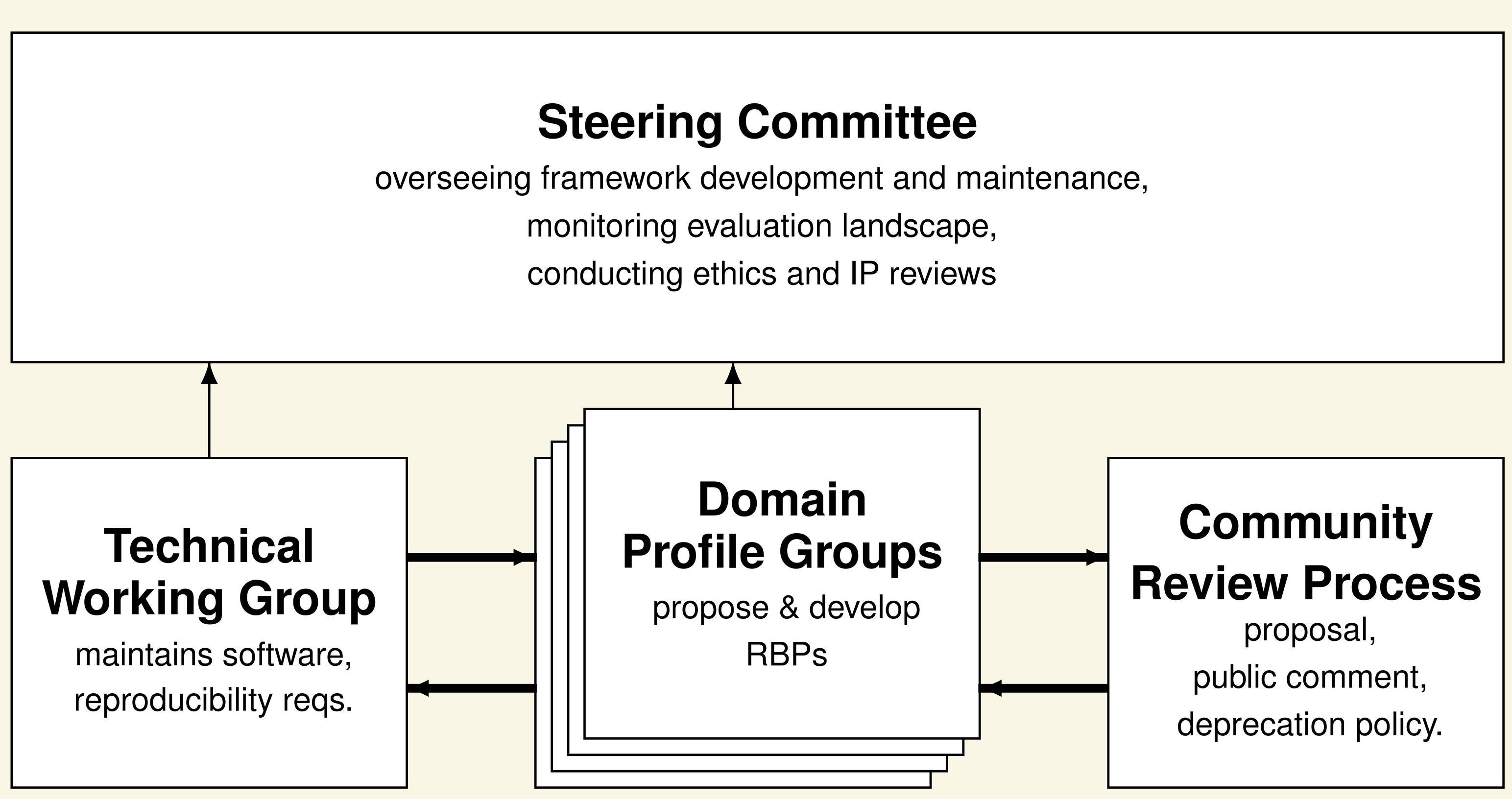

Figure: Proposed governance structure

Beyond Music

- modular architecture ⇒ easily transferable to other domains of generative AI by **replacing feature representations**; candidates are
  - *Visual art*
  - *Text generation*
  - *Game content generation*
  - *Video generation*

Framework Architecture

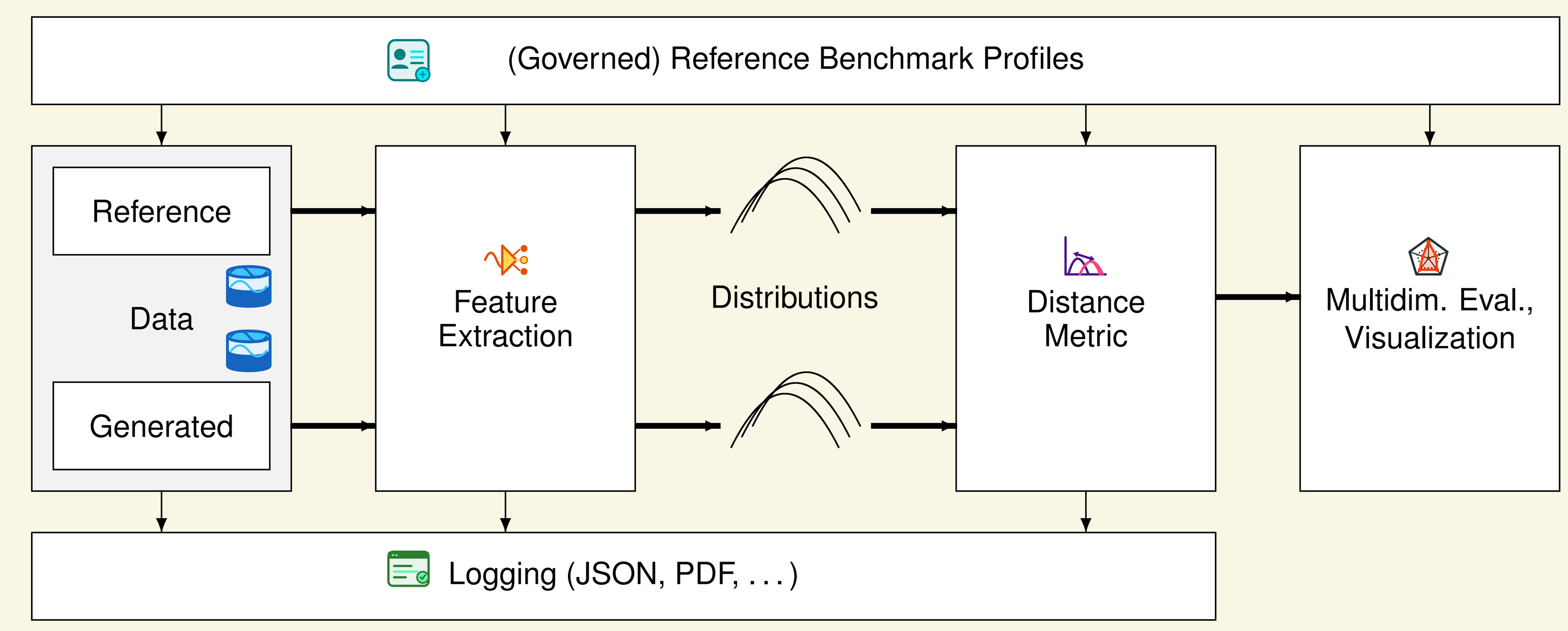

Figure: Proposed framework architecture for the evaluation of generative music.

- **Reference Benchmark Profiles (RBPs):** versioned evaluation configurations specifying
  - *reference data*
  - *feature representations*
  - *distance metrics*
  - *reporting format*
- **Standardized reference data:** publicly available, documented datasets as controllable open references.
- **Standardized feature representations:** for musical qualities, audio fidelity, controllability, and originality.
- **Standardized distance metrics:**
  - distributional distance metrics (Wasserstein, KL, MMD, etc.)
  - traditional evaluation metrics (accuracy, AUC, F1, etc.) for classification tasks
- **Multi-dimensional results & visualization:**

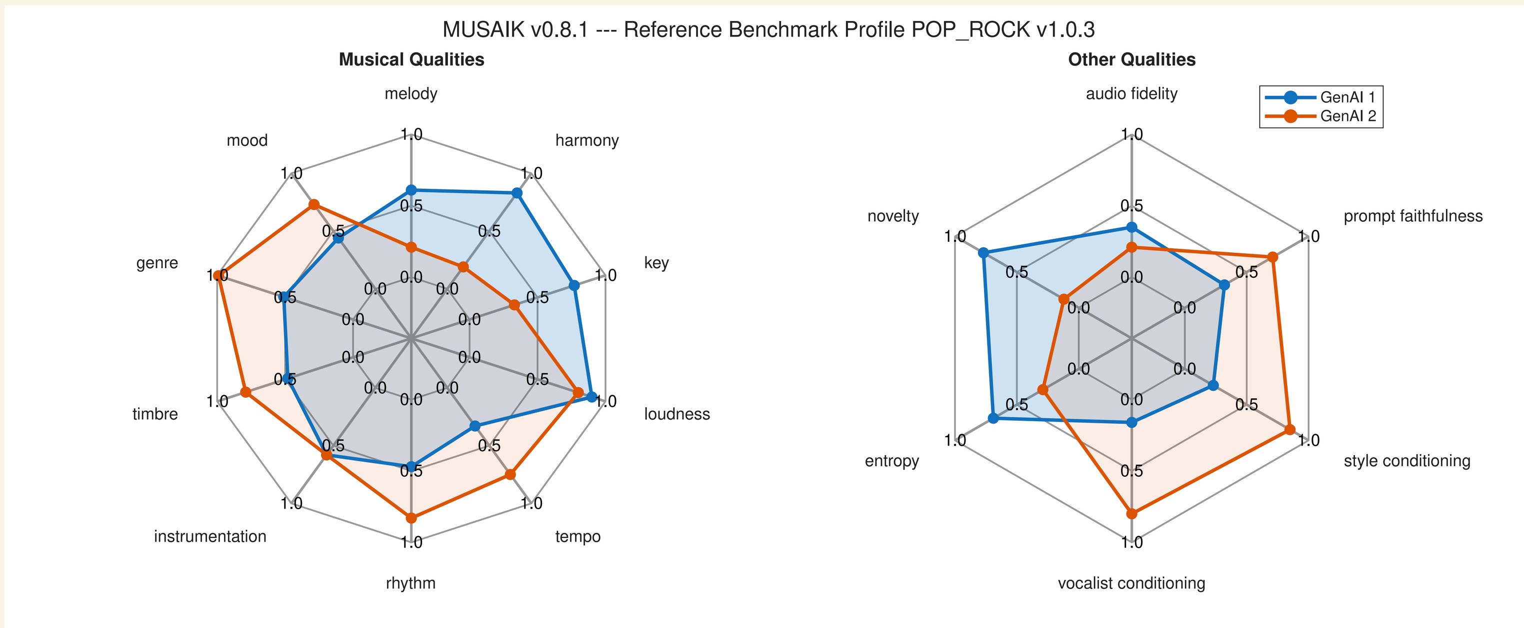

Figure: Mockup of aggregated result presentation.

- **Logging:** auditable logs (machine & human readable formats) with RBP, version info, test data, etc.

Limitations

- Missing **subjective evaluation protocols**
- **Black-box evaluation**:
  - no evaluation of the underlying model architecture/capabilities
  - no distinction between human contributions and fully machine-generated output
- **Focus on output qualities**:
  - no usability or user experience evaluation
  - no evaluation of other characteristics, such as explainability, bias, resource usage, ethical use of data, etc.

References

Alexander Lerch, Claire Arthur, Nick Bryan-Kinns, Corey Ford, Qianyi Sun, and Ashvala Vinay.
Survey on the Evaluation of Generative Models in Music.
ACM Computing Surveys, 58(4):99:1–99:36, October 2025.